Syllabus Details

Syllabus ID:	13137
Syllabus Name:	**Mathematics for Engineering_Toán cho ngành kỹ thuật**
Course Name English:	
Subject Code:	**MAE101**
Learning-Teaching Method:	Lecture,Practice
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45 contact hours (60 sessions) + 1 hour final exam + 104 hours self-study
Pre-Requisite:	None
Description:	Upon finishing the course, students must acquire: 1. the following knowledge: (ABET a1) • The basic concepts of single variable calculus: limit, derivative, integral. • Concepts of linear system of equations, matrix and their applications • The concepts of vector spaces, basis and dimension, linear transformations and their applications. • The range of application of calculus and algebra in science, technology, economics & finance... 2. the following skills: (ABET a2) • Find limit, derivative and integral of functions of single variables; • Find solutions to linear systems by using row-reduced echelon form, Cramer's Rule, inverse matrix • Find determinants and inverse of a matrix • Find linear transformations in R^2 and R^3. • Find dimensions and bases of vector spaces . • Apply the concepts of calculus and algebra in solving problems in science, technology, economics & finance, ...
StudentTasks:	Students must attend at least 80% of contact hours in order to be accepted to the final examination.
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

12 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Calculus 1						True	False	True	https://openstax.org/details/bo volume-1
2	Calculus 2						True	False	True	https://openstax.org/details/bo volume-2
3	powerpoint in CD + Powerpoint prepared by previous lecturers (PP).						False	False	False	
4	Student Video Skillbuilder in CD (Video).						False	False	False	
5	Animation in CD (Animation).						False	False	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
6	Computerized tools (for example Maxima http://maxima.sourceforge.net/) (CT)						False	False	False	
7	Essential Calculus	James Stewart	Thomson Brooks/Cole			978-0-495-10860-3	False	False	False	
8	Calculus for engineers	Donald Trim	Prentice Hall,			0-13-085603-7	False	False	False	
9	Calculus	Stanley I. Grossman				0-03-096420-2	False	False	False	
10	Linear Algebra with applications	W.Keith Nicholson,	Mc Graw Hill Education			987-1-259-07299-4.	False	True	False	
11	Linear Algebra with Applications	W. Keith Nicholson	Lyryx Online Resources	2021	2021-A		True	False	True	https://lyryx.com/linear-algebra-applications/
12	Coursera - FPT University Private Authored Courses: Linear transformation with python (Guided Project)						False	False	True	Optional practice

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe basic properties and find the limit of a function LO1.1 Find the domain, range, zeros and analyze the symmetry of the graph; create new functions from old functions. LO1.2 Find the slope and write an equation of a line; describe how a function changes under graph translations. LO1.3 Compute the limit of a function using graphs and limit laws, including the Squeeze's Theorem. LO1.4 Find limits at infinity and infinity limits, and apply in finding asymptotes of a function. LO1.5 Identify where a function is continuous; use the Intermedia Value Theorem to show that an equation has a solution.
2	CLO2	Calculate derivatives of a function LO2.1 Find the average rate of change; identify derivative as the limit of a difference quotient. LO2.2 Recognize derivative as the instantaneous velocity and the slope of tangent lines. LO2.3 Find derivatives using basic formulas and the sum, difference, product, quotient, chain rules.
3	CLO3	Use derivatives to analyze a function and to solve applications LO3.1 Use implicit differentiation to find derivatives and to solve related rates problems. LO3.2 Find linear approximation of a function; calculate differentials and apply in calculating errors of approximation. LO3.3 Apply Rolle's Theorem and Mean Value Theorem to a function; find critical numbers and absolute extrema of a function. LO3.4 Identify local extrema, and describe the monotonicity of a function; solve optimization problems. LO3.5 Recognize acceleration as second derivative; determine inflection points and describe the concavity of a function. LO3.6 Use Newton method to approximate solutions of an equation.
4	CLO4	Calculate integrals and find area under the graph of a function LO4.1 Use Riemann sums to approximate area; recognize area as a definite integral which is the limit of Riemann sums. LO4.2 Apply the Fundamental Theorems of Calculus to find derivatives of integrals, and to evaluate definite integrals. LO4.3 Recognize indefinite integrals as anti-derivatives; find integrals using basic formulas and the techniques of substitution and integration by parts. LO4.4 Use integration to solve applications and to compute the average value of a function. LO4.5 Approximate definite integrals using Midpoint, Trapezoidal and Simpson rules. LO4.6 Determine if an improper integral converges.
5	CLO5	Solve a system of linear equations using Gaussian elimination LO5.1 Identify if a matrix is in echelon forms; carry a matrix into an echelon using row operations; calculate the rank of a matrix. LO5.2 Use echelon matrices to solve/describe the solution set of a linear system.
6	CLO6	Perform basic operations on matrices and manipulate matrix transformations LO6.1 Perform addition, scalar multiplication and transpose of matrices; verify if a matrix is symmetric. LO6.2 Multiply two matrices of appropriate sizes; transform between a linear system and a matrix equation. LO6.3 Determine the matrix of a linear transformation; use properties of linear transformations in solving problems.

No.	CLO Name	CLO Details
7	CLO7	Find the determinant, inverse and diagonalization of a matrix LO7.1 Compute the cofactors and the determinant of a matrix; analyze determinants in relation with row/column operations. LO7.2 Check if a matrix is invertible; find the adjugate and the inverse of a matrix; use properties of inverses and determinants in solving problems. LO7.3 Solve a system of linear equations using inverse matrices or Cramer's rule. LO7.4 Find the characteristic polynomial, eigenvalues and eigenvectors to diagonalize a matrix; check if a matrix is diagonalizable.
8	CLO8	Manipulate vector operations and linear transformations in R^3. LO8.1 Manipulate vector operations; find equations of lines. LO8.2 Compute the dot product and the angle between two vectors; find the projection of a vector on another vector and manipulate equations of planes. LO8.3 Compute the cross product; analyze areas and volumns in relation with dot and cross products. LO8.4 Determine linear transformations in R^3 and discuss applications in computer graphic.
9	CLO9	Find the dimension and bases of a subspace LO9.1 Determine if a subset of vectors is a subspace; describe the subspace spanned by a set of vectors. LO9.2 Check if a set of vectors are independent; determine if a set of vectors form a basis. LO9.3 Find a basis and the dimension of a subspace; analyze the rank in relation with the dimension of the column, row and the null space of a matrix. LO9.4 Compute dot product and length of vectors; check if a set of vectors are orthogonal or orthonormal; write a vector as a linear combination of an orthogonal basis.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	0 Introduction. 1.1 Review of Functions. 1.2 Basic Classes of Functions	Offline	LO1	ITU	Textbook Chapter 1 (Calculus Vol. 1)		Exercises Chapter 1 (Calculus Vol. 1)	
2	2.1 A Preview of Calculus. 2.2 The Limit of a Functions	Offline	LO1	TU	Textbook Chapter 2 (Calculus Vol. 1)		Exercises Chapter 2 (Calculus Vol. 1)	
3	2.3 The Limit Laws 2.4 Continuity	Offline	LO1	TU	Textbook Chapter 2 (Calculus Vol. 1)		Exercises Chapter 2 (Calculus Vol. 1)	
4	3.1 Defining the Derivative 3.2 The Derivative as a Function	Offline	LO2	IT	Textbook Chapter 3 (Calculus Vol. 1)		Exercises Chapter 3 (Calculus Vol. 1)	
5	3.3 Differentiation Rule. 3.4 Derivatives as Rates of Change 3.5 Derivatives of Trigonometric Functions	Offline	LO2	TU	Textbook Chapter 3 (Calculus Vol. 1)		Exercises Chapter 3 (Calculus Vol. 1)	
6	3.6 The Chain Rule. 3.8 Implicit Differentiation	Offline	LO2	TU	Textbook Chapter 3 (Calculus Vol. 1)		Exercises Chapter 3 (Calculus Vol. 1)	
7	3.9 Derivatives of Exponential and Logarithmic Functions	Offline	LO2	TU	Textbook Chapter 3 (Calculus Vol. 1)		Exercises Chapter 3 (Calculus Vol. 1)	
8	4.1 Related Rates	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
9	4.2 Linear Approximations and Differentials	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
10	4.3 Maxima and Minima 4.4 The Mean Value Theorem	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
11	4.5 Derivatives and the Shape of a Graph 4.6 Limits at Infinity and Asymptotes	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
12	4.7 Applied Optimization Problems	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
13	4.9 Newton's Method	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
14	4.10 Antiderivatives	Offline	LO3	TU	Textbook Chapter 4 (Calculus Vol. 1)		Exercises Chapter 4 (Calculus Vol. 1)	
15	1.1 Approximating Areas	Offline	LO4	IT	Textbook Chapter 1 (Calculus Vol. 2)		Exercises Chapter 1 (Calculus Vol. 2)	
16	1.2 The Definite Integral 1.3 The Fundamental Theorem of Calculus	Offline	LO4	IT	Textbook Chapter 1 (Calculus Vol. 2)		Exercises Chapter 1 (Calculus Vol. 2)	
17	1.4 Integration Formulas and the Net Change Theorem	Offline	LO4	TU	Textbook Chapter 1 (Calculus Vol. 2)		Exercises Chapter 1 (Calculus Vol. 2)	
18	1.5 Substitution	Offline	LO4	TU	Textbook Chapter 1 (Calculus Vol. 2)		Exercises Chapter 1 (Calculus Vol. 2)	
19	3.1 Integration by Parts 3.6 Numerical Integration	Offline	LO4	TU	Textbook Chapter 3 (Calculus Vol. 2)		Exercises Chapter 3 (Calculus Vol. 2)	
20	3.7 Improper integrals	Offline	LO4	TU	Textbook Chapter 3 (Calculus Vol. 2)		Exercises Chapter 3 (Calculus Vol. 2)	
21	Review	Offline	LO1, LO2, LO3, LO4	U				
22	Progress Test 1	Offline	LO1, LO2, LO3, LO4	U				
23	1.1 Solutions of Systems of Linear Equations	Offline	LO5	IT	Textbook Chapter 1 (Linear Algebra)		Exercises Chapter 1 (Linear Algebra)	
24	1.1 Elementary Operations	Offline	LO5	TU	Textbook Chapter 1 (Linear Algebra)		Exercises Chapter 1 (Linear Algebra)	
25	1.2 Gaussian Elimination	Offline	LO5	TU	Textbook Chapter 1 (Linear Algebra)		Exercises Chapter 1 (Linear Algebra)	
26	1.3 Homogeneous Equations	Offline	LO5	TU	Textbook Chapter 1 (Linear Algebra)		Exercises Chapter 1 (Linear Algebra)	
27	2.1 Matrix Addition, Scalar Multiplication, and Transposition	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	
28	2.2 Matrix-Vector Multiplication	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	
29	2.2 Matrix Transformations	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
30	2.3 Matrix multiplication	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	
31	2.4 Matrix Inverses	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	
32	2.4 Properties of Inverses	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra)	
33	2.6 Linear Transformations	Offline	LO6	TU	Textbook Chapter 2 (Linear Algebra)		Exercises Chapter 2 (Linear Algebra) Optional Practice: Coursera - FPT University Private Authored Courses: Linear transformation with Python (Guided Project)	
34	3.1 The Cofactor Expansion	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
35	3.1 Properties of Determinants	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
36	3.2 Determinants and Matrix Inverses	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
37	3.2 Cramer's Rule	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
38	3.3 Eigenvalues and Eigenvectors	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
39	3.3 Diagonalization	Offline	LO7	TU	Textbook Chapter 3 (Linear Algebra)		Exercises Chapter 3 (Linear Algebra)	
40	Review	Offline	LO5, LO6, LO7	U				
41	Progress Test 2	Offline	LO5, LO6, LO7	U				
42	4.1 Vectors	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
43	4.1 Lines in Space	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
44	4.2 Projections	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
45	4.2 Planes	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
46	4.2 Cross Product	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
47	4.3 More on the Cross Product	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
48	4.4 Linear Operations on R^3	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
49	4.4 Transformations of Areas and Volumes	Offline	LO8	TU	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
50	4.5 An application in Computer Graphics	Offline	LO8	IT	Textbook Chapter 4 (Linear Algebra)		Textbook Chapter 4 (Linear Algebra)	
51	5.1 Subspaces of R^n	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
52	5.1 Spanning Sets	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
53	5.2 Linear Independence	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
54	5.2 Dimension	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
55	5.3 Dot Product, Length, and Distance	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
56	5.3 Orthogonal Sets and the Expansion Theorem	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
57	5.4 Rank of a Matrix	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
58	5.4 Rank, Nullity, and Linear Systems	Offline	LO9	TU	Textbook Chapter 5 (Linear Algebra)		Exercises Chapter 5 (Linear Algebra)	
59	Review	Offline	LO8, LO9	U				
60	Progress Test 3	Offline	LO8, LO9	U				

168 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1_DN	Discuss four ways of representing a function. Illustrate your discussion with examples.
2	1	CT_CQ1	Basis functions and the transformations of functions
3	1	HCM_CQ1	What is a function? Let's give an example. Then, find its domain and range.
4	2	CQ1_DN	Describe several ways in which a limit can fail to exist. Illustrate with sketches.
5	2	CT_CQ1	Explain the relationship between one-sided and two-sided limits.
6	2	HCM_CQ2	Suppose that the graph of f is given. Describe how the graph of the function y=f(x+a) + b can be obtained from the graph of f(x). (where a, b>0)

No.	Session No	Name	Details
7	3	CQ1_DN	What does it mean for f to be continuous at a, continuous on an interval? What can you say about the graph of such a function?
8	3	CT_CQ1	Describe three kinds of discontinuities.
9	3	HCM_CQ3	What does it mean to say that a line is the vertical asymptote of the curve? Draw curves to illustrate the various possibilities.
10	3	HCM_CQ4	What is the graph of a function? How can you tell whether a given curve is the graph of a function?
11	4	CQ1_DN	Define the derivative f'(a) . Discuss two ways of interpreting this number.
12	4	CT_CQ1	Describe three conditions for when a function does not have a derivative. Explain the meaning of a higher-order derivative.
13	4	HCM_CQ5	Describe some cases where a function is not differentiable. Illustrate with sketches.
14	5	CQ1_DN	Give several examples of how the derivative can be interpreted as a rate of change in physics, chemistry, biology, economics, or other sciences.
15	5	CT_CQ1	Predict the future population from the present value and the population growth rate.
16	5	HCM_CQ6	If y=f(x) and x varies from a to b, write expressions for the average rate of change of y with respect to x over the interval [a,b].
17	6	CQ1_DN	Find the equation of the tangent line to the folium of Descartes with equation $2x^3 + 2y^3 - 9xy = 0$ at the point (2, 1).
18	6	CT_CQ1	Recognize the chain rule for a composition of three or more functions.
19	6	HCM_CQ7	What is an implicit function?
20	6	HCM_CQ8	Explain with examples how implicit differentiation works.
21	7	CT_CQ1	Use logarithmic differentiation to determine the derivative of a function.
22	7	HCM_CQ9	How are derivatives of exponential and logarithmic functions used in real life?
23	8	CQ1_DN	How to solve a related rates problem? Illustrate with examples.
24	8	CT_CQ1	Use the chain rule to find the rate of change of one quantity that depends on the rate of change of other quantities.
25	8	HCM_CQ10	What is an example of a related rate in reality?
26	9	CQ1_DN	How to use differentials to approximate the change in the value of the function that results from a small change in the value of the input?
27	9	CT_CQ1	Describe the linear approximation to a function at a point
28	9	HCM_CQ11	What is the linearization of an everywhere differential function f?
29	9	HCM_CQ12	Suppose that f"(x) < 0 for x near a point. Is the linearization of f at the point an over-approximation or an under-approximation? Why?
30	10	CQ1_DN1	Explain the difference between an absolute maximum and a local maximum. Illustrate with a sketch.
31	10	CQ2_DN1	State the Mean Value Theorem and give a geometric interpretation.
32	10	CT_CQ1	Define absolute extrema. Explain how to find the critical points of a function over a closed interval. Provide examples.
33	10	HCM_CQ13	If f(x) is continuous on a closed interval, then it is enough to look at the points where f'(x) = 0 in order to find its absolute maxima and minima. Be prepared to justify your answer.
34	11	CT_CQ1	Use concavity and inflection points to explain how the sign of the second derivative affects the shape of a function's graph.
35	11	HCM_CQ14	What is the vertical/horizontal asymptote of a curve? Illustrate the possibilities by drawing.
36	11	HCM_CQ15	How many horizontal asymptotes can a function have in maximum?
37	12	CT_CQ1	Set up and solve optimization problems in several applied fields
38	12	HCM_CQ16	How do you solve applied optimization problems? Let's give an example.
39	13	CQ1_DN	(a) Given an initial approximation x_1 to a root of the equation f(x)=0, explain geometrically, with a diagram, how the second approximation x_2 in Newton's method is obtained. (b) Write an expression for x_2, x_n
40	13	CT_CQ1	Describe the steps of Newton's method. Recognize when Newton's method does not work. Provide examples.

No.	Session No	Name	Details
41	13	HCM_CQ17	What is Newton's method? Give examples.
42	13	HCM_CQ18	Does Newton's method always work?
43	14	CQ1_DN	A metal storage tank with volume V is to be constructed in the shape of a right circular cylinder surmounted by a hemisphere. What dimensions will require the least amount of metal?
44	14	CT_CQ1	Use antidifferentiation to solve simple initial-value problems
45	14	HCM_CQ19	What is the antiderivative? Why do we need to find the antiderivative of an integrable function?
46	15	CQ1_DN	Use six rectangles to find estimates of each type for the area under the graph of f(x)= e^(x^2) from x=0 to x=1 . (a) L_6(sample points are left endpoints) (b) R_6(sample points are right endpoints)
47	15	CT_CQ1	Compute the left and right Riemann sums—L6 and R6. Provide examples.
48	15	HCM_CQ20	How do you approximate the area under a curve? Let's take an example of the function f(x) = x^2 in the interval [1,2].
49	16	CQ1_DN	Write the definition of the definite integral of a continuous function from x=a to x=b .
50	16	CT_CQ1	Use Riemann sums to the approximate area.
51	16	HCM_CQ21	What are definite integrals? Give examples.
52	16	HCM_CQ22	What is the relationship between integrals and anti-derivatives?
53	17	CQ1_DN	Use the Net Change Theorem to evaluate the net change in function . Illustrate with examples.
54	17	CT_CQ1	Explain the significance of the net change theorem.
55	17	HCM_CQ23	How can we use definite integrals to determine the net change of a quantity?
56	18	CQ1_DN	How do you use the Substitution Rule?
57	18	CT_CQ1	When we use the substitution rule and how it is used.
58	18	HCM_CQ24	What is the substitution method for calculating integrals?
59	18	HCM_CQ25	Give an example of computing integrals by using substitution rule.
60	19	CQ1_DN	Use Midpoint Rule and Trapezoidal Rule to approximate the value of an definite integral.
61	19	CQ2_DN1	Approximate the value of an definite integral by using the Simpson’s Rule.
62	19	CT_CQ1	When we use the integrate by parts rule and how it is used.
63	19	HCM_CQ26	How to prove the integral by parts formula?
64	20	CT_CQ1	Evaluate an integral over an infinite interval.
65	20	HCM_CQ27	What are improper integrals?
66	20	HCM_CQ28	What are the real-life applications of improper integrals?
67	21	HCM_CQ29	What are the types of improper integrals?
68	22	HCM_CQ30	What is numerical integration with an example? Why do we implement numerical integration?
69	23	CQ1_DN	How many solutions can a system of linear equations have?
70	23	CT_CQ1	Show that a system consisting of exactly one linear equation can have no solution, one solution, or infinitely many solutions. Give examples
71	23	HCM_CQ31	Give example of solving a system of linear equations in real life.
72	23	HCM_CQ32	How many solutions can a system of linear equations have?
73	24	CQ1_DN	What are elementary row operations? Illustrate with examples.
74	24	CT_CQ1	By examining the possible positions of lines in the plane, show that two equations in two variables can have zero, one, or infinitely many solutions.
75	24	HCM_CQ33	What are elementary operations in linear algebra? Let's give an example.
76	24	HCM_CQ34	What are reduced row echelon matrices?
77	25	CQ1_DN	How do you solve a linear equation using the Gaussian elimination method?
78	25	CT_CQ1	Using a series of elementary row operations to carry a matrix to row–echelon form, find the rank of a matrix.
79	25	HCM_CQ35	How does Gaussian elimination work?

No.	Session No	Name	Details
80	26	CT_CQ1	Condition for a homogeneous system to have a nontrivial solution
81	26	HCM_CQ36	What are homogeneous systems?
82	27	CQ1_DN	How to find the result of Matrix Addition, Scalar Multiplication, Transposition.
83	27	CT_CQ1	Practicing some matrix operations
84	27	HCM_CQ37	How do we define addition, scalar multiplication, and transpose operations of matrices?
85	27	HCM_CQ38	State at least 2 properties of matrix addition.
86	28	CT_CQ1	What is the condition to multiply the matrix by the vector?
87	28	HCM_CQ39	Can we do the multiplication of any two matrices together?
88	28	HCM_CQ40	State at least 2 properties of matrix multiplication.
89	29	CQ1_DN	How to use matrix -vector multiplication as Matrix Transformation in R^2.
90	29	CT_CQ1	What is matrix transformation? For example.
91	29	HCM_CQ41	What is the 'matrix transformation'?
92	29	HCM_CQ42	What is the matrix of reflection on the axis Ox in the Oxy-plane?
93	30	CQ1_DN	How to multiply two matrices? Illustrate with examples.
94	30	CT_CQ1	How to multiply two matrices? For example. Is it possible to multiply any two matrices together?
95	31	CT_CQ1	Find matrix inverse. Using matrix transformations to reveal the geometrical meaning of matrix multiplication and inverse.
96	31	HCM_CQ43	When is a square matrix invertible?
97	31	HCM_CQ44	Suppose A is an invertible matrix, how to find the inverse of A?
98	32	CT_CQ1	State the properties of inverse matrices, providing examples.
99	32	HCM_CQ45	What are the properties for the inverse existence of a matrix?
100	33	CT_CQ1	what are linear transformations? State the properties of linear transformations, with examples.
101	34	CT_CQ1	How to find The Cofactor Expansion
102	34	HCM_CQ46	How to find the (i,j) -cofactor of a matrix?
103	34	HCM_CQ47	What are (i,j)-cofactors used for?
104	35	CT_CQ1	How to calculate the determinant and its properties.
105	35	HCM_CQ48	What is the determinant of a square matrix?
106	36	CT_CQ1	State the relationship formulas between the determinant of a matrix and the inverse matrix.
107	36	HCM_CQ49	Can you find determinant of non square matrices?
108	36	HCM_CQ50	Give an example of computing the determinant of a matrix by using elementary operations.
109	37	CQ1_DN	How to solve a linear system using Cramer's rule.
110	37	CT_CQ1	How can you use Cramer’s Rule to tell if a system of linear equations has one, zero, or infinitely many solutions?
111	37	HCM_CQ51	State the Cramer's Rule.
112	38	CQ1_DN	How to find the characteristic equation, the eigenvalues, and bases for the eigenspaces of the matrix.
113	38	CT_CQ1	Find the characteristic polynomial, eigenvalues and eigenvectors of matrix A. Provide examples.
114	38	HCM_CQ52	What is the characteristic polynomial of a matrix?
115	38	HCM_CQ53	What do we call the root of the characteristic polynomial of a matrix?
116	39	CT_CQ1	When is a matrix diagonalizable, give an example.
117	40	HCM_CQ54	How to find eigenvalues of a 2×2 matrix? When is it diagonalizable for sure?
118	40	HCM_CQ55	When do we guarantee a matrix is diagonalizable?
119	41	HCM_CQ56	What are the applications of diagonalization of a matrix?
120	42	CT_CQ1	What is the vector space R^n? for example

No.	Session No	Name	Details
121	42	HCM_CQ57	What is the vector space R^n?
122	43	CT_CQ1	Find the equations of the line. For example.
123	43	HCM_CQ58	How to determine the equation of a line in R^3?
124	43	HCM_CQ59	Give the vector version of the equation of a line in R^3.
125	44	CT_CQ1	Calculate the distance from the point P to the line in each case and find the point Q on the line closest to P. P(3, 2, -1) line: x = 2 +t, y = 1-t, z = 3 - 2t.
126	44	HCM_CQ60	Let U and V be vectors in R^n and V ≠ 0. What is the projection of U onto V?
127	45	CT_CQ1	Find equation planes.
128	45	HCM_CQ61	How to define a plane in R^3?
129	45	HCM_CQ62	Give the vector version of the equation of a plane in R^3.
130	46	CT_CQ1	Show that if the diagonals of a parallelogram are perpendicular, it is necessarily a rhombus.
131	46	HCM_CQ63	What is the major difference between cross product and dot product?
132	47	CT_CQ1	Find the area of the triangle.
133	47	HCM_CQ64	Why do we use cross products?
134	48	CT_CQ1	Find the matrix of the operator (reflection, projection, rotation) in R^2/ R^3.
135	48	HCM_CQ65	Give an example of a linear transformation from R^3 to R^2.
136	48	HCM_CQ66	Give an example of a linear transformation from R^3 to R^2 and indicate the matrix of this transformation.
137	49	CT_CQ1	Find the matrix of the rotation in R^3 about the x axis through the angle θ (from the positive y axis to the positive z axis).
138	49	HCM_CQ67	How is the determinant related to volume?
139	50	CT_CQ1	What are the applications of transformation in computer graphics? For example.
140	50	HCM_CQ68	What are the applications of transformation in computer graphics?
141	50	HCM_CQ69	What is the projection matrix of an orthogonal projection along a line?
142	51	CT_CQ1	When is U a subspace in R^3. For example.
143	51	HCM_CQ70	How to determine whether a subset is a subspace of R^n?
144	51	HCM_CQ71	Is R^n a subspace of itself?
145	52	CT_CQ1	How to find a spanning set of a subspace?
146	52	HCM_CQ72	How to find a finite spanning set of a subspace?
147	53	CQ1_DN	How do you tell if a vector is linearly independent or dependent?
148	53	CT_CQ1	How to find a basis of R^4.
149	53	HCM_CQ73	What is linear dependence/ linear independence of a set of vectors?
150	54	CQ1_DN	How do you find the basis and dimension of a subspace?
151	54	CT_CQ1	Calculate the dimension of R^4.
152	54	HCM_CQ74	What is the dimension of R^n?
153	54	HCM_CQ75	How to define the dimension of a subspace of R^n?
154	55	CT_CQ1	Find the length of the vector, provide example.
155	55	HCM_CQ76	How is the distance between two vectors related to dot product?
156	56	CQ1_DN	What are orthogonal and orthonormal sets?
157	56	CT_CQ1	Find the condition for the sets is orthogonal
158	56	HCM_CQ77	What are orthogonal and orthonormal sets?
159	56	HCM_CQ78	What is the main benefit of working with orthonormal sets?
160	57	CQ1_DN	how to find the rank of a matrix?
161	57	CT_CQ1	Show that the rank does not change when an elementary row or column operation is performed on a matrix.

No.	Session No	Name	Details
162	57	HCM_CQ79	How to determine the rank of a matrix in the determinant method?
163	58	CT_CQ1	Find the relationship between rank of matrix and nullity
164	58	HCM_CQ80	What is the rank and nullity of a linear transformation?
165	58	HCM_CQ81	State some relationship between the rank and the nullity of a linear transformation.
166	59	HCM_CQ82	What are positive definite matrices? Give an example.
167	59	HCM_CQ83	If A, B are positive definite matrices, is so A + B?
168	60	HCM_CQ84	List some applications of calculus and linear algebra in real life.

3 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide
1	Assignments/Exercises	on-going	3	30.0%	>0	3 hours	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	OP1: Open // OP2: (For Constructivism Approach only): Follow lecturer's proposal	OP1: At least 3/chapter // OP2: (For Constructivism Approach only): Follow lecturer's proposal	Assignment 1 covers Chapters 1, 2, 3, 4 (Vol.1), 1, 3 (Vol.2) - Calculus ; Assignment 2 covers Chapters 1, 2, 3 (Linear algebra) ; Assignment 3 covers Chapters 4, 5 (Linear algebra)	OP1: Each assignment is divided into two parts: the take-home component counts for no more than 50% of the total grade, and the in-class component counts for at least 50% (this may take the form of a presentation, essay, problem-solving, etc.). OP2: (For Constructivism Approach only): Follow lecturer's proposal

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide
2	Progress Test	on-going	3	30.0%	>0	OP1: Test 1: 40'; Test 2&3: 30' // OP2: (For Constructivism Approach only): Follow lecturer's proposal	LO1, LO2, LO3, LO4, LO5, LO6, LO7, LO8, LO9	OP1: MC or a suitable format // OP2: (For Constructivism Approach only): Follow lecturer's proposal	OP1: Test 1: 20 Test 2&3: 15+ At least 5*n questions where n is the number of Chapters covered in the test.+ Structure:• level 1 (knowledge & comprehension): 20%; • level 2 (application & analysis): 60%; • level 3 (synthesis & evaluation): 20% // OP2: (For Constructivism Approach only): Follow lecturer's proposal	Test 1 covers Chapters 1, 2, 3, 4 (Vol.1), 1, 3 (Vol.2) - Calculus ; Test 2 covers Chapters 1, 2, 3 (Linear algebra) ; Test 3 covers Chapters 4, 5 (Linear algebra)	OP1: by instructor, and by computer OP2: (For Constructivism Approach only): Follow lecturer's proposal
3	Final Exam	Final exam	1	40.0%	4	60'		Computer gradable	50 Structure: • level 1: 10% • level 2: 80% • level 3: 10%	All studied chapters. Each studied Chapter contributes 2-8 questions, depending on the number of topics in the Chapter.	by Exam Board